Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 3 ~ BookStream #002.01

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moving on to chapter 3 the high road to octave inference and so recall that the high road is like the Y so Ali please begin just go as long as you want on through the high road to octave inference okay so the high road to active inference uh begin with uh the question um about about the conditions for the Persistence of quote-unquote things I mean uh what would we expect for a thing to behave if it persists through time and that's why sometimes fap is referred to as the theory of every space thing and it's not absolutely everything but think just in this sense of uh I mean persisting through time and uh the notion that allows for this a thing to act as it as it persists through time is uh markup blankets so markup blanket is one of the most uh essential ingredients of the high road approach to active inference and that's why this section opens up with describing how Market blanket allows to describe the situation of Interest through this conceptual and mathematical tool but one thing that I always point out in almost all our all our textbook cohorts is in this section it doesn't um I mean exactly describe a markup blankets rigorously enough in terms of it's carving up the um I mean State space and so on so one thing I I've I think is can be uh clarifying in following uh all the discussions around Marco blanket is to keep in mind that it's just a boundary in the state space so it's not necessarily spatial temporal boundary although in many cases it manifests itself as spatial temporal boundaries such as the I don't know cell membranes or organelle organelle membranes and so on but it's not necessarily the case so I think it it can be helpful to keep that in mind when we every everywhere we see markup blanket uh but uh in a very very simply markup blanket uh is what allows for the agent or the system to be statistically separated from its environment as is shown here uh in figure 3.1 so basically markup blankets mathematically is just uh the uh I mean the sensory and active States together it consists of both of those States and it is statistically separates what happened externally or internally from each other so in other words internal States cannot observe or infer anything about the external States directly but only through the markov's Markov blanket and that's why it's essential to uh describe the systems in this way because obviously in any sparse coupled systems there isn't a possibility to observe the external States directly but only through those sensory and active states and and one other thing that I also believe can be a bit confusing in this picture is that yes typo in the active Estates and sensory States because it's important to observe that active state only or the flow in the active States more precisely only depends on internal and Market States or blanket States and on the other hand in sensory sensory states only depend on the external and blanket States so mu and X in those two equations should be exchanged uh yes yeah let me just unpack into this before we continue to add before we continue yes so this is what's known as the particular partition and that terminology was brought to the fore by Carl friston's famous monograph in 2019 and it's a little bit of a pun it's a particular partition because this is just one way to partition agent from environment figure from ground and we call what it partitions out the blankets and internal States as the particle so we can think about the most inert least cognitive particle is like a speck of dust doing Brownian diffusion but also there are more sophisticated kinds of cognitive particles that can include World models counterfactuals what would happen if I did this all of those kinds of sophisticated cognitive operations that we can explore in active inference so note the small typo that Ali mentioned internal States and also these edges reflect causal possibilities amongst internal external and blanket States blanket constituting the active U and sensory y States so these are map not territory this is not a spatial temporal articulation these are like causal maps of the world that may have to do with spatial temporal boundaries but are not simply that and sensory data or sensory States flow onto internal States internal states are involved in action selection resulting in active states which have some causal consequence in the world the generative process the external states which results in different sensory States coming in again and so the Markov blanket is what makes the internal and the external States conditionally independent and that can be thought of as just a no telekinesis no telepathy Clause the only way that information comes across internal and external states which are symmetrical with each other is through the boundary or the holographic screen or the Markov blanket you can continue on Ali uh okay so the next section is about surprise minimization and self-evidency so again this term self-evidencing is a common term an active inference literature first proposed by Jacob Howie so uh basically it refers to how a system or agent or in other words a particular State can gather the evidence for itself persistence or self-existence through time in other words by inferring the state of the environment and comparing that inference with its internal States or in other words with its generative model it it's basically a way I mean an interpretation of that um that kind of inference is uh on the on the one hand there is this dynamical interaction between the internal and external state but we can somehow interpret that uh physical Dynamics as engaging in the active in the active inference or as a model for um the Persistence of the agent uh through time so that's basically what refers to as self-evidencing and then we go through the subsection 3.3.1 which is which sets the stage for the recent formulation of active inference uh as in mechanics which is to uh somehow to to interpret the the active surprise minimization as minimizing the action through the hamiltonian principle of least action which is a variational principle uh and by variational principle is what what is meant here is just a computational or mathematical tool that allows for the computation or the derivative the I mean specific derivations to happen so it's not uh identical to scientific theories or scientific I don't know facts or observations it's just a tool a principle a mathematical Tool uh so again one of the main misconceptions about free energy principle is that it is on falsifiable uh obviously if we see it in this way uh it doesn't make sense to say that a mathematical tool or principle is unfalsifiable because it doesn't say anything about the empirical evidence of the phenomena we're talking about so that's why here it's important to understand how the surprise minimization can be seen as this kind of variational principle and then equation equation 3.1 draws the parallel between the surprisal as defined in section in chapter two sorry and this chapter so it kind of ties up all the arguments provided in the previous chapter with this one and how everything comes together in a single formulation of active inference that but they're just the two distinct approaches to arrive at same destination uh uh and then we go uh also sorry the other important equation and again another key equation here is equation 3.2 which is a kind of EX sorry yes equation I meant to say equation 3.2 not equation 3.1 sorry my bad so yes the other section is about the relations between inference cognition and stochastic Dynamics as uh I mean all the previous discussions around uh how the active inference can be seen as a kind of self-evidencing through the variational principles such as fep comes together to do to reframe the previous discussions we saw in Chapter 2 about perception as inference and action as inference and to see the concepts of variational free energy and expected free energy through the lens of variational free energy variational principle of least action so uh it unifies nicely all the material from uh chapters two and three into uh something that I mean not necessarily distinct from each other but just two sides of awesome yeah I find table three one to be very exciting it draws together statistical physics Bayesian information information Theory and cognitive interpretations so it's kind of like learn one thing learn many things statistical physics has been talking about minimization of variational free energy for a long time and such methods are absolutely everyday and professionalized in Bayesian statistics active inference is using exactly just that to describe perception and action and so on so that's very exciting box 3.2 describes free energy in statistical physics and active inference I sometimes joke that we have a few kinds of free energy we have Tesla like electrical power should be available to everybody for no cost we're not talking about that right now there's Gibbs free energy which is the quantity that makes chemical or thermochemical reactions irreversible like the hydrolysis of ATP and when we're talking about variational free energy we're talking about an information geometric space with similar Dynamics similar kinetics and thermodynamics but rather than describing the reaction coordinates of a chemical reaction we're thinking about it in terms of Bayesian updating and this is all called the Bayesian mechanics um 3.41 continues on with variational free energy 3.42 goes into expected free energy so we see this A Lot F variational free energy that's the real-time unfolding sensory flow and then G expected free energy and the policy planning as inference section 3.5 concludes with a novel Foundation Act of inference to understand behavior and cognition and it describes a few features of active inference including its distinguishing features from a few other ways that people have looked at behavior in cybernetics section 3.6 goes into a bit more detail on models policies and trajectories having to do with agency and policy selection 3.7 reconciliation of inactive cybernetic and predictive theories under active inference again the exact kind of thing that there's a whole literature on and it's always amazing to hear everybody's perspective on in the textbook groups 3 8 active inference from the emergence of life to agency and three nine summary you want to just give any other thoughts that you have on chapter three again it's one of the uh probably most fundamental chapters and I mean the topics covered in this chapter is essential are absolutely essential to understand everything active in Prince related but specifically the discussions in section 3.7 3.6 onward and again I believe uh is really important to understand the the broader context of active inference and how it relates to uh all the other theories but specifically section 3.8 provides a nice um view of how we can use the same modeling to the same theoretical tools to model both non-living dynamical systems or sparse couple dynamical stochastic systems and also to uh the systems that as agency or sentience so it's a much broader theoretical framework that doesn't restrict itself to only particular kinds of systems or agents and we'll see much more elaboration on that in the later wonderful all right that concludes our overview on chapter three